

1 Method

1.1 Data Collection

1.1.1 Trait selection

We selected traits in the context of three hypotheses. Traits selected under the predation satiation hypothesis included seed weight, fruit size, oil content, dispersal mode and seed dormancy, as these traits are associated with seed attractiveness to predators and the potential for surviving in seed caches. Traits selected under the pollination coupling hypothesis included pollination mode and reproductive type, as these traits are related to pollination efficiency. Traits selected under the resource matching hypothesis included drought tolerance and leaf longevity, as these traits are linked to the ability to adjust resource use under varying environmental conditions.

1.1.2 Data scraping

For our analyses, we obtained the full species list from the two volumes of *Silvics of North America*^{2,3} (hereafter referred as *Silvics*). We extracted masting data primarily from *Silvics*, with MASTREE+⁵ used as a supplementary source when there was no description on the seed production patterns in *Silvics*. Pollination mode and seed dispersal data were also obtained mainly from *Silvics*. Seed weight data were collected from the Seed Information Database by the Society for Ecological Restoration¹⁵. Seed dormancy data were gathered from *Seeds* by Baskin and Baskin¹. Leaf longevity and oil content data were accessed through the TRY database⁸. Fruit size and drought tolerance data were compiled from multiple sources, including: *Silvics*, the USDA Fire Effects Information System¹⁶, the USA National Phenology Network¹⁷, and The Gymnosperm Database⁴.

For masting data, instead of using the more commonly used coefficient of variation (CV)¹⁰, we defined masting based on qualitative description of seed production patterns. If the description said the species "have a good crop almost every year", we considered this species a non-masting species, whereas if the description said the species "have a good crop every several years or have irregular seed production", we considered this species as a masting species.

For fruit size data, we recorded the full range from the source, and took the mean value for further analyses. For pollination mode, we categorized it into wind, animal (bird, bat, or insect), or both. For seed dispersal mode, we categorized it into biotic (animal-dispersed), abiotic (wind-, gravity-, or water-dispersed), or both. For drought tolerance, we categorized it into low, moderate, or high. For leaf longevity, we standardized it to years.

In total, our dataset included 188 species, comprising 124 angiosperm species and 64 gymnosperm species. Among angiosperms, 79% of species had masting data available, and masting species accounted for 52%. Among gymnosperms, 95% of species had masting data available, and 89% species are masting species.

1.2 Statistical Analysis

1.2.1 Masting pattern and single trait

We used `phyloglm` function in the `phylolm` package (v. 2.6.2)⁷ to analyze the relationship between masting pattern and individual traits, treating masting as a binary response variable. To construct the phylogenetic tree, we verified all species names using the `WorldFlora` package (v. 1.14-5)⁹ and updated all species names according to WCVB Backbone data (2025)¹⁸. We then constructed an ultrametric phylogenetic tree in R using the `phytools` (v. 2.4-4) and `ape` (v. 5.7-1) packages^{13,14}. We calculated the half-time using the equation $(\log(2)/\alpha)$ and compared it with the tree height (standardized to 1). This metric represents the time required for the influence of ancestral states to decrease by half.

1.2.2 Masting pattern with a multivariate analysis

Since most traits in our dataset were categorical, we selected non-metric multidimensional scaling (NMDS) for a multivariate analysis. We used the `metaMDS` function in the `vegan` package (v. 2.7-1)¹¹ and tested both $k = 2$ and $k = 3$ and selected $k = 3$ based on the lower stress score. We then plotted trait distributions on two panels (NMDS1 vs. NMDS2 and NMDS2 vs. NMDS3) to check for potential patterns.

1.2.3 Phylogenetic signal for individual traits

We calculated phylogenetic signal separately for each trait type. For binary traits, we calculated Fritz and Purvis' D-statistics using the function `phylo.d` in the `caper` package (v. 1.0.4)¹². For continuous traits, we calculated the Pagel's λ using the function `phylosig` in the `phytools` package (v. 2.4-4)¹⁴. For categorical traits, we estimated the Pagel's λ using the function `fitDiscrete` in the `geiger` package (v. 2.0.11)⁶.

References

- [1] Carol C. Baskin and Jerry M. Baskin. *Seeds: Ecology, Biogeography, and Evolution of Dormancy and Germination*. Elsevier/Academic Press, San Diego, California, 2 edition, 2014. ISBN 9780124166776.
- [2] Russell M. Burns and Barbara H. Honkala, editors. *Silvics of North America*, volume 1 of *Agriculture Handbook*. United States Department of Agriculture, Forest Service, Washington, DC, 1990. URL <https://research.fs.usda.gov/treesearch/1547>.
- [3] Russell M. Burns and Barbara H. Honkala, editors. *Silvics of North America*, volume 2 of *Agriculture Handbook*. United States Department of Agriculture, Forest Service, Washington, DC, 1990.

- [4] Christopher J. Earle. The gymnosperm database. <https://www.conifers.org/>, 2026. Accessed 2026-05-13.
- [5] Andrew Hacket-Pain, Jessie J. Foest, Ian S. Pearse, Jalene M. La-Montagne, Walter D. Koenig, Giorgio Vacchiano, Michał Bogdziewicz, Thomas Caignard, Paulina Celebias, Joep van Dormolen, et al. Mastree+: Time-series of plant reproductive effort from six continents. *Global Change Biology*, 28(9):3066–3082, 2022. doi: 10.1111/gcb.16130. URL <https://doi.org/10.1111/gcb.16130>.
- [6] Luke J. Harmon, Jason T. Weir, Chad D. Brock, Richard E. Glor, and Wendell Challenger. Geiger: investigating evolutionary radiations. *Bioinformatics*, 24(1):129–131, 2008. doi: 10.1093/bioinformatics/btm538. URL <https://CRAN.R-project.org/package=geiger>.
- [7] Lam Si Tung Ho and Cécile Ané. phylolm: Phylogenetic linear regression. *R package version*, 2(1), 2014. URL <https://CRAN.R-project.org/package=phylolm>.
- [8] Jens Kattge, Gerhard Bönsch, Sandra Díaz, Sandra Lavorel, Iain C. Prentice, Paul Leadley, Gerhard Bönsch, Eric Garnier, Mark Westoby, Peter B. Reich, Ian J. Wright, et al. Try plant trait database – enhanced coverage and open access. *Global Change Biology*, 26(1):119–188, 2020. doi: 10.1111/gcb.14904. URL <https://doi.org/10.1111/gcb.14904>.
- [9] Roeland Kindt. Worldflora: An r package for exact and fuzzy matching of plant names against the world flora online taxonomic backbone data. *Applications in Plant Sciences*, 8(9):e11388, 2020. doi: 10.1002/aps3.11388. URL <https://CRAN.R-project.org/package=WorldFlora>.
- [10] W D Koenig and J M Knops. Patterns of annual seed production by northern hemisphere trees: a global perspective. *The American Naturalist*, 155(1):59–69, 2000.
- [11] Jari Oksanen, Gavin L. Simpson, F. Guillaume Blanchet, Roeland Kindt, Pierre Legendre, Peter R. Minchin, R. B. O’Hara, Peter Solyomos, M. Henry H. Stevens, Eduard Szoecs, and Helene Wagner. vegan: Community ecology package. *R package version 2.6-8*, 2024. URL <https://CRAN.R-project.org/package=vegan>.
- [12] David Orme, Robert Freckleton, Gavin Thomas, Thomas Petzoldt, Susanne Fritz, Nick Isaac, and Will Pearse. caper: Comparative analyses of phylogenetics and evolution in r. *R package version*, 5(2), 2013. URL <https://CRAN.R-project.org/package=caper>.
- [13] Emmanuel Paradis, Julien Claude, and Korbinian Strimmer. Ape: Analyses of phylogenetics and evolution in r language. *Bioinformatics*, 20(2):289–290, 2004. doi: 10.1093/bioinformatics/btg412. URL <https://CRAN.R-project.org/package=ape>.

- [14] Liam J. Revell. *phytools: An R package for phylogenetic comparative biology (and other things)*. CRAN, 2024. URL <https://CRAN.R-project.org/package=phytools>.
- [15] Society for Ecological Restoration, International Network for Seed-based Restoration, and Royal Botanic Gardens, Kew. Seed information database (sid). <https://ser-sid.org/>, 2023. Accessed 2026-05-13.
- [16] United States Department of Agriculture, Forest Service. Fire effects information system (feis). <https://www.feis-crs.org/feis/>, 2026. Accessed 2026-05-13.
- [17] USA National Phenology Network. Usa national phenology network. <https://www.usanpn.org/>, 2026. Accessed 2026-05-13.
- [18] World Flora Online. World flora online taxonomic backbone, 2025. URL <http://www.worldfloraonline.org>. Version 2025.12. Published on the Internet; accessed 2026-05-01.